Leaf litter decomposition along an altitudinal gradient

A. Ferrante · L. Moretti · S. Bianchi

Department of Ecology, a fictional university. A sample document: the data are invented.

ABSTRACT

Decomposition of leaf litter returns most of the carbon fixed by a forest to the atmosphere, and its rate is governed chiefly by temperature and moisture. We followed the mass loss of beech litter in mesh bags placed at five elevations between 400 and 1 800 m over two years. Decomposition constants fell from 0.62 to 0.31 per year with altitude, a decline steeper than temperature alone predicts (Figure 1, Table 1). We argue that the snowpack, by insulating the litter in winter, offsets part of the cold, and that the residual decline reflects slower colonisation by soil fauna above the tree line.

1. Introduction

Mountain slopes compress, within a few kilometres, a climatic range that would otherwise span a continent. They have therefore long served as natural laboratories for asking how temperature controls the processes of an ecosystem. Litter decomposition is one of the most studied of these processes, because it is slow enough to integrate seasonal variation and fast enough to be measured within the length of a grant.

The classical expectation is a simple one. Microbial respiration roughly doubles for every ten degrees of warming, so decomposition should slow with elevation in proportion to the lapse rate of temperature. Field studies have often found a weaker dependence than this, and two explanations compete for it. The first holds that the microbial community acclimates to the local climate, so that a cold site decomposes faster than a warm community would at the same temperature. The second holds that winter matters more than the annual mean suggests: under a deep snowpack the soil stays near freezing while the air above it falls far below, and decomposition continues.

Here we test the second explanation directly, by comparing sites with and without a lasting winter snowpack across the same range of mean annual temperature. The methods are set out in the next section, the results in Results (p. 2).

2. Methods

Five sites were chosen on the north-facing slope of a single valley, at 400, 750, 1 100, 1 450 and 1 800 m. At each site we buried ten loggers recording soil temperature at a depth of 2 cm every hour, and placed sixty mesh bags, each holding 5.00 g of air-dried beech leaves collected at the lowest site in a single autumn. Bags of two mesh sizes were used: a fine mesh of 0.25 mm, which excludes soil fauna, and a coarse mesh of 5 mm, which admits them.

Ten bags of each mesh were retrieved from each site at four, eight, twelve, eighteen and twenty-four months. Their contents were cleaned of soil and roots, dried at 60 °C to constant mass and weighed. The decomposition constant k was estimated for each site and mesh by fitting the single exponential model $M(t) = M_0\,\mathrm{e}^{-kt}$ to the remaining mass by least squares.

Snow cover was recorded at each site by a time-lapse camera photographing a graduated stake once a day. A site was counted as snow-covered on a given day when the stake showed at least 10 cm of snow.

Table 1. The five sites, their climate over the two years of the study, and the decomposition constant of litter in coarse mesh bags.

Site	Elevation (m)	Mean soil temp. (°C)	Snow days per year	k (per year)
Valley floor	400	11.2	4	0.62
Lower beech	750	9.1	21	0.55
Upper beech	1 100	7.0	68	0.49
Spruce	1 450	5.3	112	0.43
Above the tree line	1 800	3.4	151	0.31

3. Results

Litter lost mass at every site throughout the study, including during the winter months. After two years the coarse mesh bags at the valley floor retained 29 % of their original mass, and those above the tree line 54 %. The decomposition constants for every site are listed in Table 1 and shown against mean soil temperature in Figure 1.

Across the four forested sites, k declined by 0.04 per year for every degree of cooling. Had the soil followed the air through the winter, the expected decline over the same range would have been close to 0.07. The difference is concentrated in the winter months: at the two highest forest sites the soil under snow stayed between −0.5 and 0.5 °C for most of the winter, while the air above it reached −15 °C.

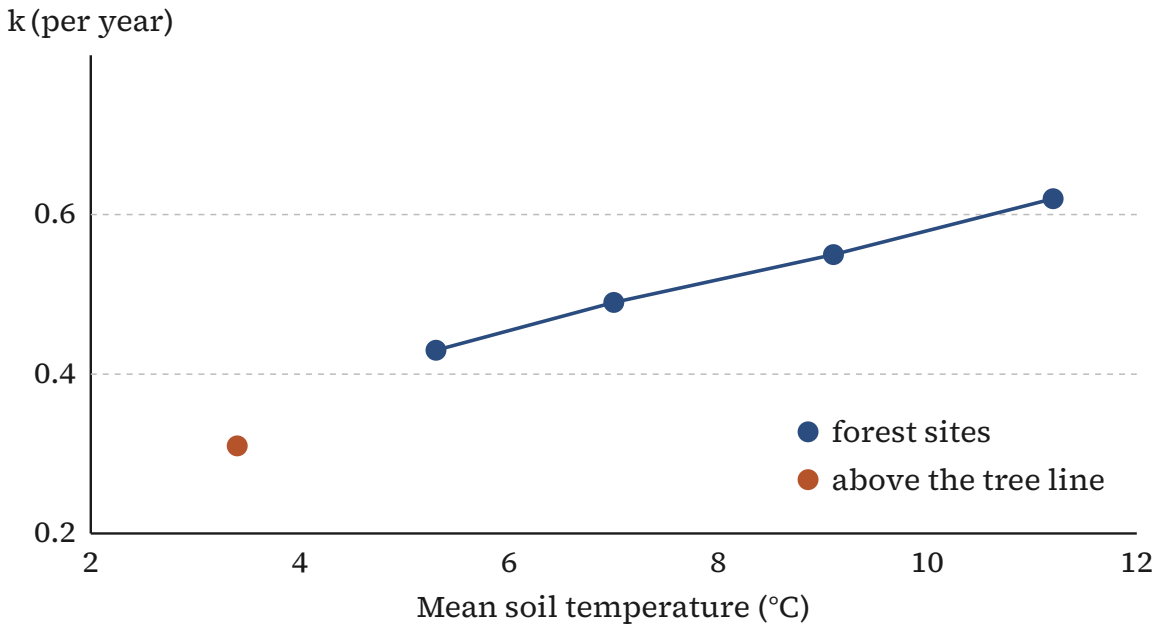


Figure 1. Decomposition constant of litter in coarse mesh bags against mean soil temperature. Blue: the four forested sites. Red: the site above the tree line, which falls well below the forest trend.

The site above the tree line departs from the forest trend. Its k of 0.31 per year is lower than the forest relationship predicts for its temperature by about a third. In fine mesh bags, which exclude soil fauna, the difference vanishes: the fine mesh constants of all five sites lie on one line against temperature.

4. Discussion

Two results stand out. The first is that decomposition is less sensitive to elevation than temperature alone predicts, and that the shortfall arises in winter. This is what the insulation hypothesis expects: where snow lies for months, the litter beneath it spends the winter close to freezing rather than far below it, and microbial activity, although slow, does not stop. The number of snow days rises steeply with elevation at our sites, so the higher the site, the more of the cold it is spared.

The second is that the alpine site is slow for a reason that has nothing to do with temperature. Its decline disappears when fauna are excluded, which points to the soil animals that fragment litter and carry it into the soil. Earthworms, which do most of this work in the beech forest, were absent from every sample taken above the tree line. Their absence, rather than the climate, accounts for the third of decomposition the alpine site lacks.

Both results caution against predicting the response of mountain ecosystems to warming from temperature alone. A warmer climate with less snow could, paradoxically, slow winter decomposition at middle elevations, while the upward movement of soil fauna could speed it above the present tree line far more than the warming itself.

References

1. Olson, J. S. Energy storage and the balance of producers and decomposers in ecological systems. *Ecology* 44, 322–331 (1963).
2. Swift, M. J., Heal, O. W. & Anderson, J. M. *Decomposition in Terrestrial Ecosystems* (Blackwell, Oxford, 1979).
3. Hobbie, S. E. Temperature and plant species control over litter decomposition in Alaskan tundra. *Ecological Monographs* 66, 503–522 (1996).
4. Bradford, M. A. et al. Understanding the dominant controls on litter decomposition. *Journal of Ecology* 104, 229–238 (2016).